

Rhombus — Answers

Final answers with a one-line reason. For full methods, see the worked examples in the notes.

Objective

Q1 (c) **Rhombus**. All four sides equal; a square is a special rhombus.

Q2 (b) **Rhombus**. Diagonals bisecting at right angles $\Rightarrow$ rhombus.

Q3 **90**. Rhombus diagonals are perpendicular.

Q4 (a) **50°**. Opposite angles are equal.

Q5 **False**. They are perpendicular but generally unequal (equal only in a square).

Q6 (a) **Both true, R explains A**. Equal sides make a square a rhombus.

Short answer

Q7 **5 cm**. Half-diagonals 3 and 4: $\sqrt{3^2 + 4^2} = 5$.

Q8 **70°, 110°, 70°**. Opposite equal (110°); adjacent supplementary (70°).

Q9 **x = 90°**. Diagonals of a rhombus intersect at right angles.

Long answer

Q10 **60°, 120°, 60°, 120°**. The diagonal = side makes an equilateral triangle (all 60°); adjacent angle = 120°.

Q11 **13 cm**. Half-diagonals 5 and 12: $\sqrt{5^2 + 12^2} = 13$.

Q12 **A square**. Four equal sides make it a rhombus; one 90° angle forces all angles to 90°, so it is a square.

Total: 21 marks.